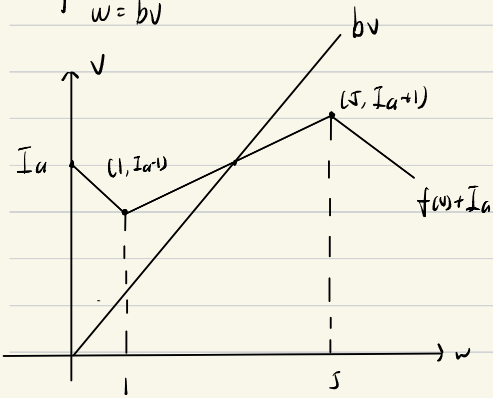
1.

$$\begin{cases} \dfrac{dv}{dt} = f(v) - w + I_a = 0 \\ \dfrac{dw}{dt} = bv - w = 0 \end{cases}$$

$$\Rightarrow \begin{cases} w = f(v) + I_a \\ w = bv \end{cases}$$




to have a periodic behaviour

$$b > \frac{I_a+1-(I_a-1)}{5-1} = \frac{1}{2}$$

$$b < I_a - 1 \quad \text{and} \quad 5b > I_a + 1$$

$$\therefore \quad b > \frac{1}{2} \quad \text{and} \quad b+1 < I_a < 5b-1$$

2. $I_a = \int y^2 \, dA(y)$

$= 4\int_1^5 y^2 \cdot 2 \cdot dy + 2\int_0^1 y^2 \cdot 12 \, dy$

$= 8\int_1^5 y^2 \, dy + 24\int_0^1 y^2 \, dy$

$= \frac{8}{3}(5^3 - 1) + 24 \cdot \frac{1}{3}$

$= 338.67 \, cm^4$

$I_B = 2\int_0^6 2y^2 \, dy + 2\int_2^4 y^2 \cdot 8 \, dy$

$= 4\int_0^6 y^2 \, dy + 16\int_2^4 y^2 \, dy$

$= 4 \times \frac{6^3}{3} + \frac{16}{3} \cdot (4^3 - 2^3)$

$= \frac{1760}{3} \, cm^4 = 586.67 \, cm^4$

$\therefore I_B > I_A$

$\therefore$ B will resist the bending better.